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GMAT math formulas: the cheat sheet you actually need.

Every formula tested on GMAT Focus Quant, organized by topic with a one-line when-to-use note for each. No calculator and no formula sheet on test day — so the high-frequency ones have to live in your head.

The GMAT hands you nothing. On the Quantitative section there is no on-screen calculator and there is no formula sheet, on any section, ever. Whatever you need to compute, you compute by hand, and whatever formula you need, you supply from memory. That changes what a cheat sheet is for. This is not a document to glance at during the exam — you can't. It is a drilling tool. Read it, cover a column, recite the other, and keep doing that until recall is automatic. The goal is to make the formula appear in your head the instant you recognize the question type.

> A formula you can look up is useless on test day, because you can't look anything up. The only formulas that count are the ones you can produce cold, under the clock, with no prompt.

Quant on the Focus Edition is Problem Solving only: 21 questions in 45 minutes, multiple choice. Data Sufficiency now lives in Data Insights. The formulas below cover the full Quant content map. Each table pairs the formula with a one-line note on when to reach for it, because knowing a formula and knowing *when* it applies

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and bleed into the word problems, so fluency here pays off across the whole section.

Formula	Use it when…
Percent change = (new − old) / old × 100	a value goes up or down and you need the percentage, not the raw difference. Always divide by the *original* value.
Increase by p% → multiply by (1 + p/100); decrease → multiply by (1 − p/100)	chaining successive changes. A 20% rise then a 20% fall is ×1.2×0.8 = 0.96, not back to start.
x% of y = (x/100) × y = y% of x	translating "percent of" into arithmetic; the swap trick makes some calculations far easier.
Fraction ↔ decimal ↔ percent equivalents	you want to skip long division. Know 1/8 = 0.125 = 12.5%, 1/6 ≈ 16.7%, 1/3 ≈ 33.3%, and the rest of the common set on sight.
Ratio a : b means parts; total = a + b parts	a quantity is split in a given ratio. Scale every part by the same multiplier to recover actual amounts.

Number properties

High-frequency and almost never requiring heavy computation — these reward structure over arithmetic. Prime factorization is the master key to most of them.

Rule / formula	Use it when…
Divisibility: by 3 if digit sum divisible by 3; by 9 if digit sum by 9; by 4 if last two digits by 4; by 6 if even and divisible by 3	testing factors fast without dividing. Combine the rules for composite divisors.

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4, 8, 6)	power. Find the cycle length, then take the exponent mod that length.
Remainder form: n = dk + r, with 0 ≤ r < d	reasoning about remainders. The remainder r is what matters; d is the divisor and k is the quotient.

Algebra

Less about exotic formulas and more about recognizing patterns instantly. The factoring identities below show up constantly, often disguised, and spotting them turns a grind into one line.

Formula	Use it when...
Linear: solve ax + b = c → x = (c − b)/a	one unknown, one equation. For two unknowns, use substitution or elimination across two equations.
Quadratic formula: x = (−b ± √(b² − 4ac)) / 2a	a quadratic won't factor cleanly. The discriminant b² − 4ac tells you how many real roots exist.
Difference of squares: a² − b² = (a + b)(a − b)	you see two squares subtracted. One of the highest-value patterns on the test; it collapses ugly expressions fast.
Perfect squares: (a ± b)² = a² ± 2ab + b²	expanding a binomial square, or recognizing one to factor in reverse. Watch the middle 2ab term.
FOIL: (a + b)(c + d) = ac + ad + bc + bd	multiplying any two binomials by hand. First, Outer, Inner, Last — the systematic way to avoid dropping a term.

Exponents and roots

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x^{n+m}; $x^n / x^m = x^{n-m}$	exponents; never multiply the bases.
Power of a power: $(x^n)^m = x^{nm}$	an exponent is raised to another exponent. Multiply them.
Negative exponent: $x^{-n} = 1 / x^n$	you want to clear a negative power. Flip the base to the denominator and make the exponent positive.
Fractional exponent: $x^{m/n} = \sqrt[n]{(x^m)}$	a root is written as a power, or vice versa. The denominator is the root, the numerator is the power.
Root rules: $\sqrt{(ab)} = \sqrt{a} \cdot \sqrt{b}$; $\sqrt{(a/b)} = \sqrt{a} / \sqrt{b}$	simplifying radicals. Pull perfect-square factors out from under the root.

Word problems

The translation tax. The arithmetic is rarely hard; converting a sentence into the right equation is where points are won and lost. These four templates cover the overwhelming majority of Quant word problems.

Formula	Use it when...
Distance = rate × time (D = RT)	anything moves. Build a small table of distance, rate, and time for each traveler, then relate them.
Combined work rate: 1/a + 1/b = 1/t (rates add)	two or more agents work together. Add their per-unit rates, not their times, to get the shared rate.
Weighted average = Σ(value × weight) / Σ weights	mixing groups of different sizes, or blending solutions of different concentrations. The bigger group pulls the average toward it.
Simple interest: I = Prt.	a question specifies simple vs. compound. Simple grows on principal only; compound grows on principal plus prior

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Two full readings from the Verbal Foundations chapter — the same material paying students get, with Quant and Data Insights samples one click away.

Read the free sample chapter →

Statistics

The GMAT loves descriptive stats because they reward understanding over calculation. You will almost never compute a standard deviation by hand — but you must know what it measures and how a data shift affects it.

Formula / concept	Use it when…
Mean = sum of values / count	the question says “average.” Often you're given the mean and count and must back out the sum.
Median = middle value (or average of the two middle)	data is skewed or has outliers. Order the set first; the median ignores how extreme the tails are.
Range = greatest − least	you need a quick measure of total spread. It only sees the two endpoints.
Standard deviation = spread around the mean (conceptual)	comparing how tightly clustered two sets are. Adding a constant to every value leaves SD unchanged; multiplying scales it.
Evenly spaced set: sum = average × count; average = (first + last)/2	summing a consecutive or arithmetic sequence. Pairing the ends gives the average instantly.
	counting terms in an evenly spaced list. The “+ 1” is

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Counting and probability

reduces to a few ideas: multiply independent choices, decide whether order matters, and count the complement when direct counting is messy.

Formula	Use it when…
Fundamental counting principle: total = $n_1 \times n_2 \times \ldots$	independent stages each have their own number of choices. Multiply across the stages.
Permutations: nPr = n! / (n − r)!	order matters — arrangements, rankings, seatings. ABC and CBA count as different outcomes.
Combinations: nCr = n! / [r!(n − r)!]	order does *not* matter — choosing a group or committee. ABC and CBA are the same selection.
Probability = desired outcomes / total outcomes	outcomes are equally likely. Count the favorable cases and the total cases, then divide.
At least one: P(at least one) = 1 − P(none)	"at least one" appears and direct counting is ugly. Computing the complement is almost always faster.

Geometry

The longest list, because geometry is formula-dense. The figures are usually not drawn to scale, so you reason from given values and these relationships, not from how the picture looks.

Formula	Use it when…
Angles: triangle sums to 180°; straight line 180°; full circle 360°	any figure with unknown angles. Chase angles using these sums plus vertical and parallel-line rules.
Triangle area = ½ × base × height	you have a base and the perpendicular height. The height must be perpendicular to the chosen base, not

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	hypotenuse is a leg times √2.
30-60-90 sides in ratio 1 : √3 : 2	a 30-60-90 triangle, or half of an equilateral. The side opposite 30° is the smallest.
Circle: circumference = 2πr; area = πr²	any circle problem. Diameter d = 2r; arcs and sectors are fractions of these by the central angle over 360°.
Slope = $(y_2 - y_1)/(x_2 - x_1)$	comparing line steepness or testing parallel (equal slopes) and perpendicular (negative reciprocals).
Distance = $\sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$	finding the length of a segment in the coordinate plane. It's the Pythagorean theorem in disguise.
Midpoint = $((x_1 + x_2)/2, (y_1 + y_2)/2)$	you need the point halfway between two coordinates. Average the x-values and the y-values separately.
Box: volume = lwh; surface area = 2(lw + lh + wh)	a rectangular solid. Volume fills it; surface area wraps the six faces.
Cylinder: volume = πr²h; surface area = 2πr² + 2πrh	a cylinder. The lateral surface is the circumference times the height; add the two circular caps.

What to actually memorize

You cannot brute-force this list by rote and expect it to hold under pressure. Split it in two. A small set of high-frequency formulas appears so often that hesitation costs you real time and points — those must be reflexive: **percent change**, **D = RT**, the **combined work rate**, the **30-60-90 and 45-45-90 special triangles**, and **nCr**. Drill those until they fire without conscious effort.

Everything else, you derive from understanding rather than store as a fact. If you understand that area is base times height for a parallelogram, the triangle's half follows. If you understand FOIL, the perfect-square and difference-of-squares

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The platform

Zakarian GMAT's Quant chapters teach each of these formulas in context, then immediately put them to work in graded problem sets so the recall is built through retrieval rather than re-reading. The error log's tags surface whether a miss was a formula gap or a misapplication, which tells you exactly which row of this cheat sheet to drill next. The sample chapter is free if you want to see the teaching first.

Read the free Quant sample chapter →

Read the Quant timing strategy

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